

Step 1: Find the last digit

The last digit of a number can be found by using the modulo operator (`%`). The expression $n \% 10$ will give the remainder when n is divided by 10, which is the last digit.

Step 2: Find the first digit

To find the first digit, repeatedly divide the number by 10 until it becomes a single-digit number.

Step 3: Write the C code

C



```
#include <stdio.h>
int main() {
    int n, firstDigit, lastDigit;
    printf("Enter a number: ");
    scanf("%d", &n);
    lastDigit = n % 10;
    firstDigit = n;
    while (firstDigit >= 10) {
        firstDigit = firstDigit / 10;
    }
    printf("First digit: %d\n", firstDigit);
    printf("Last digit: %d\n", lastDigit);
    return 0;
}
```

Step 1: Understand the factorial formula

The factorial of a non-negative integer n , denoted by $n!$, is the product of all positive integers less than or equal to n . The formula

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 1$$

For example, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.
factorial of 0 is 1.

Step 2: Write the C code

C

```
#include <stdio.h>
int main() {
    int n, i;
    unsigned long long factorial = 1;
    printf("Enter a non-negative integer: ");
    scanf("%d", &n);
    if (n < 0) {
        printf("Error! Factorial of negative number is not defined.\n");
    } else {
        for (i = 1; i <= n; ++i) {
            factorial *= i;
        }
        printf("Factorial of %d = %llu\n", n, factorial);
    }
    return 0;
}
```

Step 1: Understand the definition of a prime number

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

Step 2: Write the C code

C

```
#include <stdio.h>
int main() {
    int n, i, flag = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n == 0 || n == 1)
        flag = 1;
    for (i = 2; i <= n / 2; ++i) {
        if (n % i == 0) {
            flag = 1;
            break;
        }
    }
    if (flag == 0)
        printf("%d is a prime number.\n");
    else
        printf("%d is not a prime number.\n");
    return 0;
}
```

```
#include <stdio.h>

long long factorial(int n) {
    long long fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

int main() {
    int num, originalNum, digit;
    long long sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);

    originalNum = num;

    while (num > 0) {
        digit = num % 10;
        sum += factorial(digit);
        num /= 10;
    }

    if (sum == originalNum) {
        printf("%d is a strong number\n");
    } else {
        printf("%d is not a strong number\n");
    }

    return 0;
}
```


5. Find sum of all digits in a number

Step 1: Understand the process

To find the sum of all digits in a number, you can repeatedly extract the last digit using the modulo operator (%) and add it to a sum, then remove the last digit by integer division (/) until the number becomes 0.

Step 2: Write the C code

C



```
#include <stdio.h>
int main() {
    int n, temp, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
        sum += temp % 10;
        temp /= 10;
    }
    printf("Sum of digits of %d = %d\n",
    return 0;
}
```